

Education

- **Pusan National University** Busan, Republic of Korea
M.S. in Computer Engineering Mar. 2024 – Feb. 2026
Advisor: Prof. Yoon-Ho Choi
Thesis: Cloud-Native System Call Monitoring with a Distributed Sidecar Architecture
- **Pusan National University** Busan, Republic of Korea
B.S. in Computer Sciences and Engineering Mar. 2018 – Feb. 2024
Served mandatory military service May 2020 – Nov. 2021

Employment

- **Zero Trust Cloud Security Research Center** Busan, Republic of Korea
Researcher Mar. 2026
- **ERUTY** Busan, Republic of Korea
Software Engineer Dec. 2022 – Mar. 2023

Publications

- **Hybrid Session-Based UEBA for AWS Environment**
Doo-Hwan Choi, Geonwoo Yoon, Jae-Seok Kim, Yoon-Ho Choi
The 17th International Conference on Ubiquitous and Future Networks (ICUFN), 2026.
- **Container-Specific Service Mesh-based System for Mitigating Lateral Movement Attacks**
Geonwoo Yoon, Jinmyeong Shin, Jae-Seok Kim, Seunghyuk Kim, Jaeyoung Jeong, Yoon-Ho Choi
IEEE Transactions on Cloud Computing, 2026. [Paper]
- **Lightweight Service Mesh for Intrusion Detection using KD-CNN in Cloud-Native Environments**
Geonwoo Yoon, Jae-Seok Kim, Seunghyuk Kim, Jaeyoung Jeong, Millati Pratiwi, Yoon-Ho Choi
ACM Cloud Computing Security Workshop (CCSW in conjunction with the ACM CCS), 2025. [Paper] [Video]
- **Automated Cybersecurity Risk Assessment and Visualization Framework for Resilient Cloud IT Asset Management**
Joonho Seo, Jinmyeong Shin, Geonwoo Yoon, Jae-Seok Kim, Yoon-Ho Choi
ACM/SIGAPP Symposium on Applied Computing (SAC), 2025. [Paper]

Under Review

- **Caught Today, Missed Tomorrow: Inducing Detection Regression in Continual Anomaly Detection**
Geonwoo Yoon, Yoon-Ho Choi
Under review (double-blinded)
- **SCALE: Risk-aware Distributed System Call Audit Framework for Cloud-Native Environments**
Geonwoo Yoon, Seunghyuk Kim, Jaeyoung Jeong, Jae-Seok Kim, Yoon-Ho Choi
Under review (double-blinded)
- **TeRA: Closing the Cross-Container Gap with Technique-Derived Syscall Filters and Reactive Activation**
Jaeyoung Jeong, Geonwoo Yoon, Millati Pratiwi, Yoon-Ho Choi
Under review (double-blinded)

- **ADAPT: Agentic Detection-Aware Penetration Testing in Monitored Environments**
Seunghyuk Kim, Geonwoo Yoon, Yoon-Ho Choi
Under review (double-blinded)
- **MADS: Margin-Aware Divergent Switching for Moving Target Defense Against Adaptive Adversarial Attacks**
Jae-Seok Kim, Joonho Seo, Geonwoo Yoon, Yoon-Ho Choi
Under review (double-blinded)
- **Context-Preserving Adversarial System Call Injection for Cloud Anomaly Detection Evasion**
Jae-Seok Kim, Seunghyuk Kim, Geonwoo Yoon, Joonho Seo, Yoon-Ho Choi
Under review (double-blinded)

Projects

- **Online Learning-based Service Mesh for Lateral Movement Attacks**
Project Manager & Research Engineer
Developed by CSRC
Feb. 2026 – Apr. 2026
- **Enhancing VPD Control Web Framework (SCOPE)**
Developer
In collaboration with LG Electronics
Aug. 2025 – Dec. 2025
- **PNU Zero Trust CSRC CASB (Cloud Access Security Broker) [Web]**
Research Engineer
Developed by S3Lab
Jul. 2024 – May 2025
- **Industrial Coating Film Defect Detection and Monitoring System**
Research Engineer
In collaboration with GBLIGHT
Jul. 2024 – Dec. 2024
- **Web-based Psychoeducational Platform [Web]**
Project Manager & Developer
In collaboration with the Dept. of Psychology, PNU
May 2024 – Nov. 2024
- **PNU Cloud RMF (Risk Management Framework).v1**
Research Engineer
Developed by S3Lab
Dec. 2023 – May 2024

Teaching Experience

- **Operating Systems**
Teaching Assistant
Pusan National University
1st Semester 2024

Honors and Awards

- **Spear-phishing response training mini-challenge**
Dec. 2023
– Awarded **Third Place** in the spear-phishing response training mini-challenge, hosted by the Ulsan Information Security Center and Korea Internet & Security Agency (KISA).
- **4th PATH-HACK Makerthon**
Feb. 2023
– Awarded **First Place** in the PATH-HACK Makerthon for developing an integrated software–hardware prototype, hosted by Pathfinder & Korean Institute of Startup and Entrepreneurship Development.
- **Academic Excellence Scholarship**
1st & 2nd Semester 2022
- **National Science & Technology Scholarship**
1st Semester 2018
– Merit-based national STEM scholarship awarded for outstanding scores in math and science on the Korean CSAT.

Patents

- **Geonwoo Yoon**, Yoon-Ho Choi, “RISK-BASED DIFFERENTIATED DISTRIBUTED SYSTEM CALL COLLECTION FRAMEWORK DEVICE AND ITS OPERATION METHOD IN A CLOUD-NATIVE ENVIRONMENT”. Korean Patent Application No. 10-2026-0068229 (Pending)
- **Geonwoo Yoon**, Yoon-Ho Choi, “DISPATCHER-BASED DISTRIBUTED SYSTEM CALL COLLECTION DEVICE AND METHOD FOR CLOUD-NATIVE ENVIRONMENTS”. Korean Patent Application No. 10-2026-0068266 (Pending)
- **Geonwoo Yoon**, Yoon-Ho Choi, “METHOD AND SYSTEM FOR DETECTING ABNORMAL BEHAVIOR TO PREVENT LATERAL MOVEMENT ATTACKS IN KUBERNETES ENVIRONMENT”. Korean Patent Application No. 10-2025-0067413 (Pending)
- **Geonwoo Yoon**, Yoon-Ho Choi, “ZERO-COPY BASED MAIN CONTAINER’S SYSTEM CALL COLLECTION DEVICE AND METHOD FOR USING EBPF IN KUBERNETES SECONDARY CONTAINER”. Korean Patent Application No. 10-2025-0067396 (Pending)
- Jae-Seok Kim, **Geonwoo Yoon**, Yoon-Ho Choi, “EQUIPMENT CONTROL BASED VISION METHOD AND SYSTEM FOR AI-BASED DEFECT DETECTION AND DATA COLLECTION IN HIGH-SPEED AND WIDE-AREA REFLECTIVE FILMS”. Korean Patent Application No. 10-2024-0133547 (Pending)

Skills

Languages	C/C++, Python, JavaScript
Libraries	Pytorch, TensorFlow, scikit-learn, React, FastAPI
Tools	Docker, Kubernetes, eBPF, PostgreSQL, pgAdmin